

KEMCHUG, Russian Federation

WMO: 295620

Lat: 56.0936N

Lon: 91.7166E

Elev: 332

StdP: 97.40

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.9	(c) -32.6	(d) -39.0	(e) 0.1	(f) -35.6	(g) -35.7	(h) 0.1	(i) -32.4	(j) 8.6	(k) -10.6	(l) 7.5	(m) -10.2	(n) 0.7	(o) 30	(p) 0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 28.3	(d) 18.5	(e) 26.5	(f) 17.7	(g) 24.8	(h) 17.1	(i) 20.0	(j) 25.2	(k) 19.0	(l) 24.3	(m) 18.1	(n) 23.2	(o) 2.6	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.6	22.2	17.1	12.7	21.3	16.1	11.9	20.5	59.1	25.2	55.6	24.4	52.5	23.3	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.5	5.8	-38.0	31.4	4.7	1.3	-41.4	32.4	-44.2	33.2	-46.9	33.9	-50.3	34.9
			-38.1	21.8	4.7	1.4	-41.5	22.8	-44.3	23.6	-46.9	24.4	-50.3	25.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-16.5	-13.8	-5.3	2.0	9.0	16.4	17.7	15.1	8.8	1.4	-7.8	-14.2
	DBStd	13.43	8.80	7.80	6.29	5.51	5.64	4.14	3.06	3.69	4.52	5.53	7.49	8.25
	HDD10.0	3933	821	667	474	246	89	5	0	4	75	270	533	749
	HDD18.3	6339	1079	900	732	491	293	83	47	113	286	524	783	1007
	CDD10.0	705	0	0	0	5	59	196	239	162	40	4	0	0
	CDD18.3	69	0	0	0	0	4	24	28	13	1	0	0	0
	CDH23.3	694	0	0	0	1	47	269	240	123	14	0	0	0
	CDH26.7	139	0	0	0	0	9	61	44	26	1	0	0	0
Wind	WSAvg	2.5	2.4	2.6	3.0	3.0	3.0	2.1	1.9	1.9	2.3	2.7	2.8	2.6
Precipitation	PrecAvg	535	22	19	20	31	46	72	65	76	60	49	43	34
	PrecMax	657	41	39	40	55	99	137	138	161	112	79	80	58
	PrecMin	402	3	5	3	10	13	25	19	25	15	17	21	9
	PrecStd	64	9	8	9	12	22	29	30	31	26	15	12	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	2.9	11.0	19.2	27.7	30.8	30.1	29.5	25.3	17.6	9.3	1.4
		MCWB	-1.6	-0.8	3.9	9.5	15.7	18.2	20.1	20.1	16.1	10.0	5.5	-1.5
	2%	DB	-1.8	0.4	7.5	16.0	24.1	28.4	27.9	26.5	21.6	14.4	6.0	-0.7
		MCWB	-3.7	-2.3	2.3	8.2	13.6	17.8	19.1	18.6	14.3	8.6	3.1	-2.6
	5%	DB	-3.5	-1.2	5.2	13.3	21.6	26.4	26.0	24.2	18.9	11.6	3.6	-2.2
		MCWB	-5.2	-3.4	1.0	6.7	12.4	17.2	18.3	17.5	12.9	7.2	0.9	-3.7
	10%	DB	-5.1	-3.0	3.2	10.9	18.9	24.5	24.3	22.0	16.3	9.4	1.4	-3.8
		MCWB	-6.6	-4.8	-0.3	5.3	11.1	16.3	17.7	16.2	11.6	6.1	-0.5	-5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.1	-0.1	4.7	10.3	16.4	20.1	22.2	21.3	16.6	11.1	5.9	-1.0
		MCDB	0.5	2.1	9.9	17.1	26.7	25.9	27.6	27.5	22.0	16.0	8.6	0.3
	2%	WB	-3.4	-1.9	2.7	8.9	14.5	19.2	20.4	19.3	14.8	9.2	3.4	-2.3
		MCDB	-2.0	-0.1	6.7	15.2	21.8	25.6	25.6	24.8	20.3	13.4	6.0	-0.8
	5%	WB	-5.1	-3.4	1.3	7.2	13.2	18.2	19.4	18.1	13.5	7.8	1.1	-3.6
		MCDB	-3.6	-1.4	4.7	12.6	19.7	24.6	24.0	22.8	17.8	11.5	3.4	-2.2
	10%	WB	-6.7	-4.8	-0.1	5.4	11.8	17.3	18.4	16.9	12.3	6.1	-0.4	-5.3
		MCDB	-5.1	-3.0	3.0	10.4	18.0	23.1	23.0	21.0	15.6	9.0	1.1	-3.9
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.5	8.8	9.0	10.1	10.7	9.5	9.5	8.6	6.3	6.2	6.7
		MCDBR	5.6	7.4	10.6	13.0	14.2	14.4	12.8	13.3	13.3	10.2	6.4	5.6
		MCWBR	4.9	6.0	6.9	7.6	7.1	6.2	5.7	6.3	7.0	6.1	4.3	4.9
	5% WB	MCDDBR	5.3	6.8	9.4	12.2	12.3	11.8	10.5	11.5	11.5	9.7	5.9	5.7
		MCWBR	4.9	5.8	6.4	7.6	6.7	5.8	5.4	5.9	6.5	6.1	4.3	5.2
Clear-Sky Solar Irradiance	taub		0.284	0.317	0.365	0.396	0.411	0.406	0.428	0.410	0.371	0.351	0.317	0.285
	taud		2.005	1.985	1.927	2.070	2.194	2.301	2.252	2.273	2.329	2.222	2.032	1.952
	Ebn at Noon		573	707	765	813	827	837	807	796	776	676	517	450
	Edh at Noon		66	102	139	141	132	121	125	115	94	81	65	55
All-Sky Solar Radiation	RadAvg		0.78	1.75	3.10	4.31	4.94	5.92	5.30	4.23	2.74	1.42	0.70	0.47
	RadStd		0.09	0.14	0.26	0.29	0.54	0.51	0.33	0.37	0.35	0.13	0.10	0.06

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page